

Homework 6

Task 1

I first created a dataframe of the provided RData file. When looking at the prices, I realized that the difference within the brands were very small. This can be seen by the following table as well:

Brand	Price.1st		Price.3rd			
	Price.Min.	Qu.	Price.Median	Price.Mean	Qu.	Price.Max.
Price_A	3.0801068	3.0871148	3.0899857	3.0900117	3.0929485	3.0997759
Price_B	0.4900026	0.4924996	0.4948924	0.4949724	0.4975153	0.4999981
Price_Outside	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000

Notice that the minimum and maximum price for each brand are close to each other, meaning that there is not much price variation. I am not sure if this was planned, as the prices differed much more in the last homework. When I used the data simulation that was mentioned at the end of the task descriptions, I also got prices for the brands with more variance. I literally just imported the dataset and did not change anything in the data myself.

In the first plot, we can also see that the distribution is peaky for brand A and brand B. As there is a grouping into colors by brand, the plot reveals essentially already the brand-conditional price distribution.



Figure 1: Marginal Price Distribution

Task 2

Because of the huge price differences between Brand A and Brand B, it seems obvious that Brand B will be preferred by most consumers. This can also be seen by the count of overall brand preferences across all observations of all consumers (7000 total). For each consumer, we have 7 observations.

Brand	Count
Brand_A	972
Brand_B	5375
Outside_Option	653

This corresponds to the following share:

Brand	Proportion
Brand_A	0.13885714
Brand_B	0.76785714
Outside_Option	0.09328571

Generally, 220 consumers only chose one Brand across the 7 choice tasks. And in ALL cases, Brand B was the Brand that consumers chose. This suggests that the prices in the data make Brand B more attractive compared to Brand A and the outside option. Although, every consumer prefers Brand A weakly at the same price, Brand B is generally the favourite as the prices are not close to each other. Brand B is actually much closer to the outside option, as the analysis in Task 1 revealed.

Task 3

The data doesn't say anything about price sensitivity, because basically the prices are fixed for the three brands. This is obviously not great for a researcher as we cannot learn too much about price sensitivity. Generally, the data is very uninformative about brand preference and price sensitivity (if the data that I have is correct).

Task 4

For this task I wrote the script in file "Task4.R", which fits the `rmnlIndepMetrop` model from `bayesm` to the data.

Moments	mean	std dev	num se	rel eff	sam size
1	0.18	2.9	0.0144	1.1	21600

Moments	mean	std dev	num se	rel eff	sam size
2	-0.16	9.0	0.0445	1.1	21600
3	2.02	1.4	0.0071	1.1	21600

Quantiles	2.5%	5%	50%	95%	97.5%
1	-5.5	-4.61	0.19	5.0	5.9
2	-17.7	-14.92	-0.18	14.7	17.5
3	-0.8	-0.35	2.01	4.4	4.9

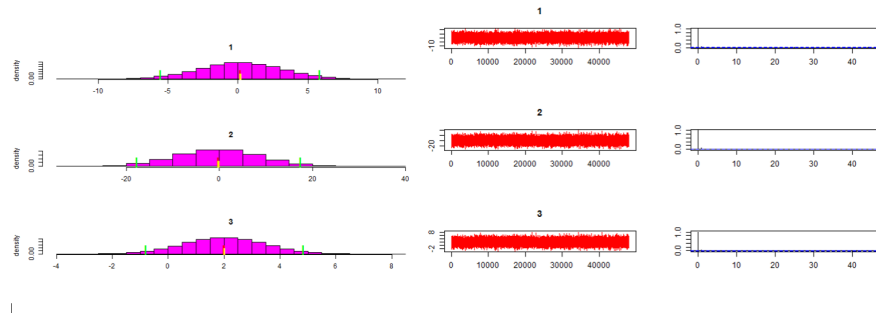


Figure 2: Posterior distribution

The coefficients are not identified well. Especially the second coefficient exhibits strong uncertainty.

Task 5

For this task, I could not get the provided code “SampleCallToRccp.R” to run. There was always the error: “Error: \$ operator is invalid for atomic vectors”, which showed up when i was just running the plain script without modifications. The issue occured when calling the “rhierMnlRwMixture_SR” function in the last line. It is likely related to the input “Data” that might have a different structure as expected. I tried to mitigate the issue by creating a new Data Variable which contains a list and add the p and the Z:

```
Data <- list(
  p = 3,
  lgtdata = E_Data,
  Z = NULL
)
```

Hierarchical Model

```
out_order = rhierMnlRwMixture_SR(Data=Data,  
                                  Prior=Prior,Mcmc=Mcmc,nvar_c=nvar_c,flag="approx")
```

But the error remained. The code “SampleCallToRccp.R” with my changes is also attached.